

BrokerVerse

Out-of-the-box insurance broking platform

Solution Architecture · Document 08 of 11

RTO and RPO

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Document control

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Review and approval

Role	Name	Signature	Date
Solution architect			
DevOps / infrastructure lead			
Information security			
Business owner			

Conventions

- Statements of fact are derived from the code, configuration and database of the baseline above (file paths are given relative to the repository). Figures measured on the local test system are labelled **test data** or **indicative**.
- Targets, sizes, retention periods and thresholds that are not facts of the code are labelled **Recommended / to be confirmed by the business and DevOps**.
- Items marked **Gap** are defects or missing capabilities found while preparing this document.

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1 Introduction

1.1 Purpose and scope

This document proposes the recovery time objective (RTO) and recovery point objective (RPO) for BrokerVerse, explains how the recommended architecture meets them for each failure scenario, and lists what must be in place and tested for the objectives to hold. Backup configuration and restore procedures are in document 09; the high-availability design in document 07.

Recommended / to be confirmed by the business and DevOps: All objectives in this document are proposals. The business owner must confirm them against the impact of an outage on premium collection, claims service and regulatory reporting, and DevOps must confirm that the provisioned services (retention, Multi-AZ, cross-region copies) and the tested restore times meet them. Nothing in the baseline code or configuration sets an RTO or RPO.

1.2 Definitions

- **RPO** (recovery point objective): the maximum period of committed data that may be lost, measured back from the incident.
- **RTO** (recovery time objective): the maximum time from the declaration of the incident until users can work again.
- **Service tiers:** groups of functions or data with the same objectives.

2 Business impact and service tiers

Tier	Functions and data	Rationale
Tier 1: core transactions	Sign-in; policies, quotations, endorsements, claims; bills, receipts, journals, vouchers, commissions, remittances, debit notes (PostgreSQL)	Financial and contractual records; loss requires re-keying from paper and bank evidence and may breach record-keeping duties
Tier 2: documents	Uploaded documents and photos, generated statements (EFS / UPLOAD_DIR)	Evidence for policies and claims; many can be re-obtained from clients or regenerated
Tier 3: operational and derived data	Notifications, e-mail outbox, generated report files, scheduler history, dashboards	Regenerated by jobs or re-run by users
Tier 4: front end and configuration as code	SPA build, backend image, migrations, seeds	Rebuilt from Git and the pipelines

3 Proposed objectives

Scenario	RPO	RTO	How the architecture meets it
API task or container failure	0	< 5 min (typically seconds)	At least two tasks behind the ALB; failed task replaced by ECS; no data held in the task
Availability Zone failure	0 (Tier 1 and 2)	< 15 min	RDS Multi-AZ synchronous standby, automatic failover in 1 to 2 minutes; EFS regional storage; tasks in the second AZ keep serving
Database instance or storage failure	0 (Multi-AZ)	< 15 min	Multi-AZ failover; without Multi-AZ: restore from PITR (RPO about 5 min, RTO 1 to 2 h)
Logical corruption or erroneous mass change (bad release, wrong purge, user error)	<= 15 min (target 5 min)	<= 4 h	PITR to a new RDS instance just before the incident, verification, switch of <code>DATABASE_URL</code> ; transactions entered after the recovery point are re-entered from the audit trail and source documents
Loss or corruption of uploaded files	<= 24 h	<= 4 h	AWS Backup of EFS (daily); files of the day after the last backup must be re-uploaded; EFS restore to a new path or file system
Regional outage (ap-southeast-1 unavailable)	<= 24 h (daily cross-region copies)	<= 24 h (warm standby: 8 h)	Restore the latest cross-region RDS snapshot and EFS recovery point in the recovery region, deploy the same images and the SPA, repoint DNS / CloudFront
Accidental deletion of the front end	0	< 1 h	S3 versioning or re-run of the pipeline on the last commit, CloudFront invalidation
Loss of secrets	n/a	< 1 h	Secrets Manager versions; sealed escrow copy of <code>DATA_ENCRYPTION_KEY</code> and <code>JWT_SECRET</code> (document 09)

Summary proposal for the business: **Tier 1 RPO <= 15 minutes (5 minutes achievable with PITR) and RTO <= 4 hours for a restore; RTO <= 15 minutes for infrastructure failures in the region; Tier 2 RPO <= 24 hours; regional disaster RPO <= 24 hours and RTO <= 24 hours.**

4 How the objectives are achieved

4.1 Recovery point

- **Amazon RDS automated backups** keep daily snapshots and the transaction log (WAL), uploaded about every five minutes, for the retention period (recommended 35 days, the RDS maximum); a point-in-time restore is possible to any second up to the latest restorable time, normally within five minutes of the present. This gives the Tier 1 RPO of 5 to 15 minutes.
- **Multi-AZ** replicates synchronously to the standby, so an instance or AZ failure loses no committed transaction (RPO 0).
- **Cross-region snapshot copies** (daily) bound the loss in a regional disaster to one day. If the business needs less, use cross-region automated backup replication (continuous, PITR in the second region) or an RDS cross-region read replica (seconds).
- **EFS** is backed up daily by AWS Backup; files are the Tier 2 RPO driver. If daily is not enough, schedule additional EFS backups (for example every 4 hours) or move documents to S3 with versioning and cross-region replication (later change).
- The application writes each business operation in one transaction, so every recovery point is transactionally consistent; e-mails and notifications created after the recovery point are simply not present.

4.2 Recovery time

- Infrastructure failures are handled automatically (ALB health checks, ECS task replacement, RDS failover) within minutes.
- Restores (PITR, snapshot) create a new RDS instance; for a database of a few GB (document 06) the restore itself takes about 20 to 60 minutes; the remaining RTO covers verification, switching `DATABASE_URL`, restarting tasks and business checks. On a restored database of the same release the API finds no pending migration, re-runs the idempotent seed and becomes ready as soon as it connects.
- Recovery in another region requires the images (ECR replication), secrets and infrastructure definitions to exist there; with infrastructure as code (recommended) the environment can be created in 2 to 4 hours, with a pre-built warm standby in under 1 hour plus data restore.

4.3 Dependencies for the objectives

Prerequisite	Status in the baseline
RDS Multi-AZ, backups 35 days, PITR	To be provisioned (checklist: daily snapshots with at least 30 days' retention)
Cross-region snapshot and EFS copies	To be provisioned
EFS for <code>UPLOAD_DIR</code> with AWS Backup	To be provisioned (checklist: EFS backup)
Secrets in Secrets Manager with escrow of <code>DATA_ENCRYPTION_KEY</code>	To be provisioned (checklist: keep the key with the backups)
Infrastructure as code (CloudFormation / Terraform / CDK)	Not in the repository; recommended to meet the regional RTO
Backend image pipeline and image retention in ECR	Not in the repository (front-end pipeline only)
Restore runbook and restore tests	Document 09; first test before go-live (checklist)
Monitoring and alerting to detect incidents quickly	Document 11; to be provisioned

5 Testing and review

Test	Frequency (recommended)	Evidence
RDS point-in-time restore to a new instance, application smoke test against it	Quarterly and before go-live	Restore duration, recovery point reached, smoke test result
Multi-AZ failover (reboot with failover) in UAT	Twice a year	Failover time, errors seen by users
EFS restore of a sample of files	Quarterly	Files readable through the application
Regional recovery exercise (tabletop yearly, full every two years)	Yearly	Achieved RTO / RPO against the objectives
Review of objectives with the business	Yearly or after major change	Signed-off objectives

6 Related documents

This document is one of eleven that together form the BrokerVerse Solution Architecture. They are maintained and issued together (source and build: [docs/architecture/README.md](#)).

No.	Document	File
01	Solution Component Diagram and Application Architecture	01_BrokerVerse_Solution_Component_Diagram_Application_Architecture.pdf
02	Database Design	02_BrokerVerse_Database_Design.pdf
03	Database Inventory	03_BrokerVerse_Database_Inventory.pdf
04	Technology Stack	04_BrokerVerse_Technology_Stack.pdf
05	Shared Service Components	05_BrokerVerse_Shared_Service_Components.pdf
06	Capacity and Performance	06_BrokerVerse_Capacity_and_Performance.pdf
07	High Availability and Resiliency	07_BrokerVerse_High_Availability_and_Resiliency.pdf
08	RTO and RPO (this document)	08_BrokerVerse_RTO_and_RPO.pdf
09	Backup and Recovery	09_BrokerVerse_Backup_and_Recovery.pdf
10	Data Archival, Housekeeping and Restoration	10_BrokerVerse_Data_Archival_Housekeeping_and_Restoration.pdf
11	Monitoring	11_BrokerVerse_Monitoring.pdf

7 Glossary

Terms and abbreviations used in this document.

Term	Meaning
ALB	Application Load Balancer (AWS): distributes HTTPS requests over the API containers and checks their health.
API	Application programming interface; here the BrokerVerse REST API under /api served by the Node.js backend.
Audit log / audit trail	Table audit_log: who changed which record, when, from which IP, with the before and after values.
AZ	Availability Zone: an isolated data-centre group inside an AWS region.
CloudFront	AWS content delivery network that serves the single-page application and can route /api to the backend.
ECR	Elastic Container Registry (AWS): stores the backend container images.
ECS / Fargate	Elastic Container Service (AWS) and its serverless compute mode for running the API containers.
EFS	Elastic File System (AWS): shared, multi-AZ network file system proposed for the upload directory.
JWT	JSON Web Token: signed token carrying the user id, roles, permissions and token version.
Migration	Numbered SQL file in backend/src/db/migrations applied once, in order, on API start.
Multi-AZ	Deployment with a synchronous standby in a second Availability Zone and automatic failover (RDS).
PITR	Point-in-time recovery: restoring a database to any second inside the backup retention window.
RDS	Amazon Relational Database Service, the managed PostgreSQL target.
RPO	Recovery point objective: the maximum acceptable data loss, measured in time.
RTO	Recovery time objective: the maximum acceptable time to restore service.
S3	Amazon Simple Storage Service: object storage hosting the SPA build (and proposed archive store).
Seed	Idempotent data load run on every API start: reference data always, sample data only when SEED_SAMPLE_DATA is on.
SPA	Single-page application: the React front end served as static files.
UAT	User acceptance testing environment.
WAL	Write-ahead log: PostgreSQL change log used for replication and point-in-time recovery.